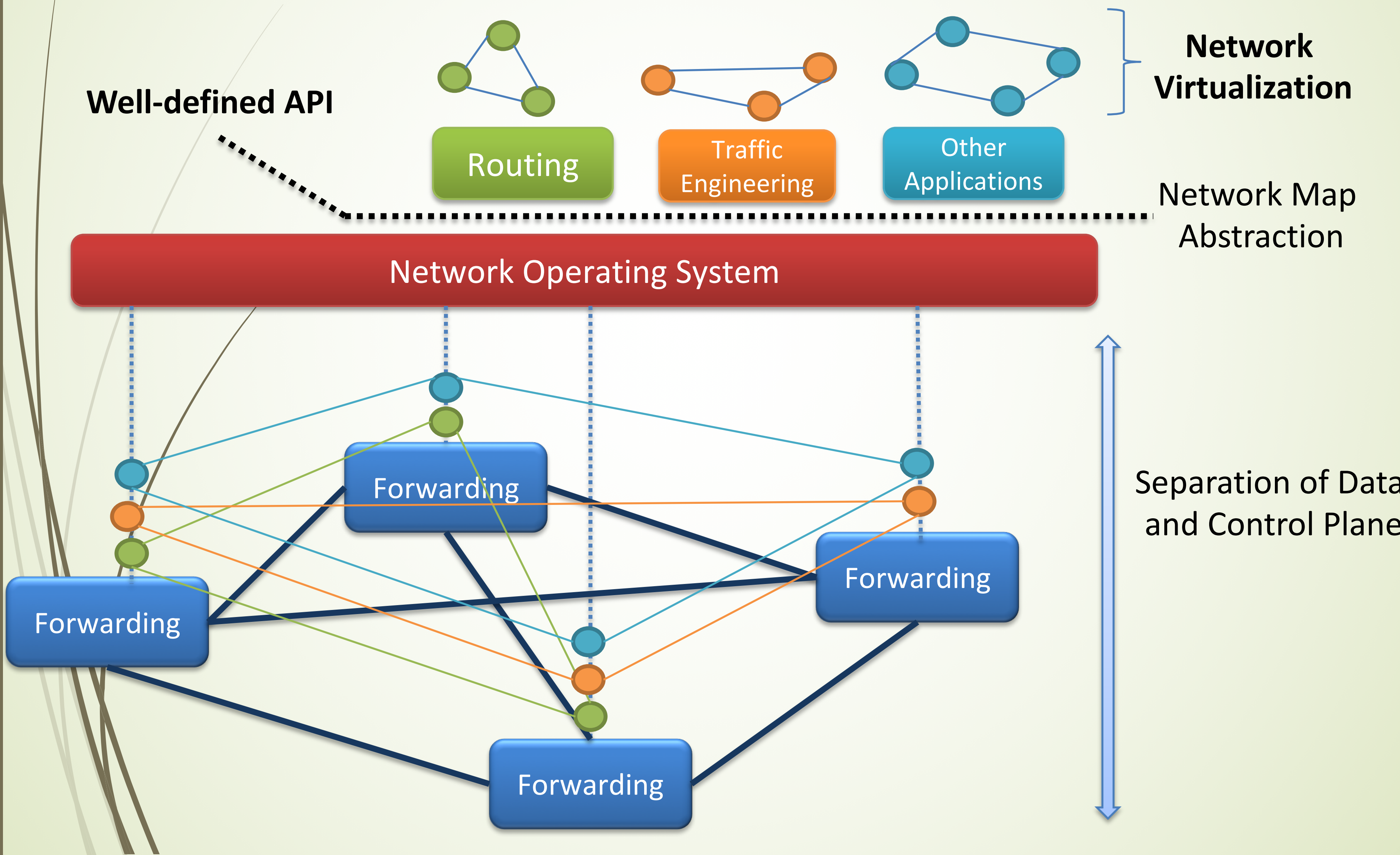
Control Plane Programs

- Control program operates on view of network
  - Input: global network view (graph/database)
  - Output: configuration of each network device
- Examples:
  - Routing
  - Traffic engineering
  - Deep packet inspection
  - etc.
- **Q: Are control programs distributed?**

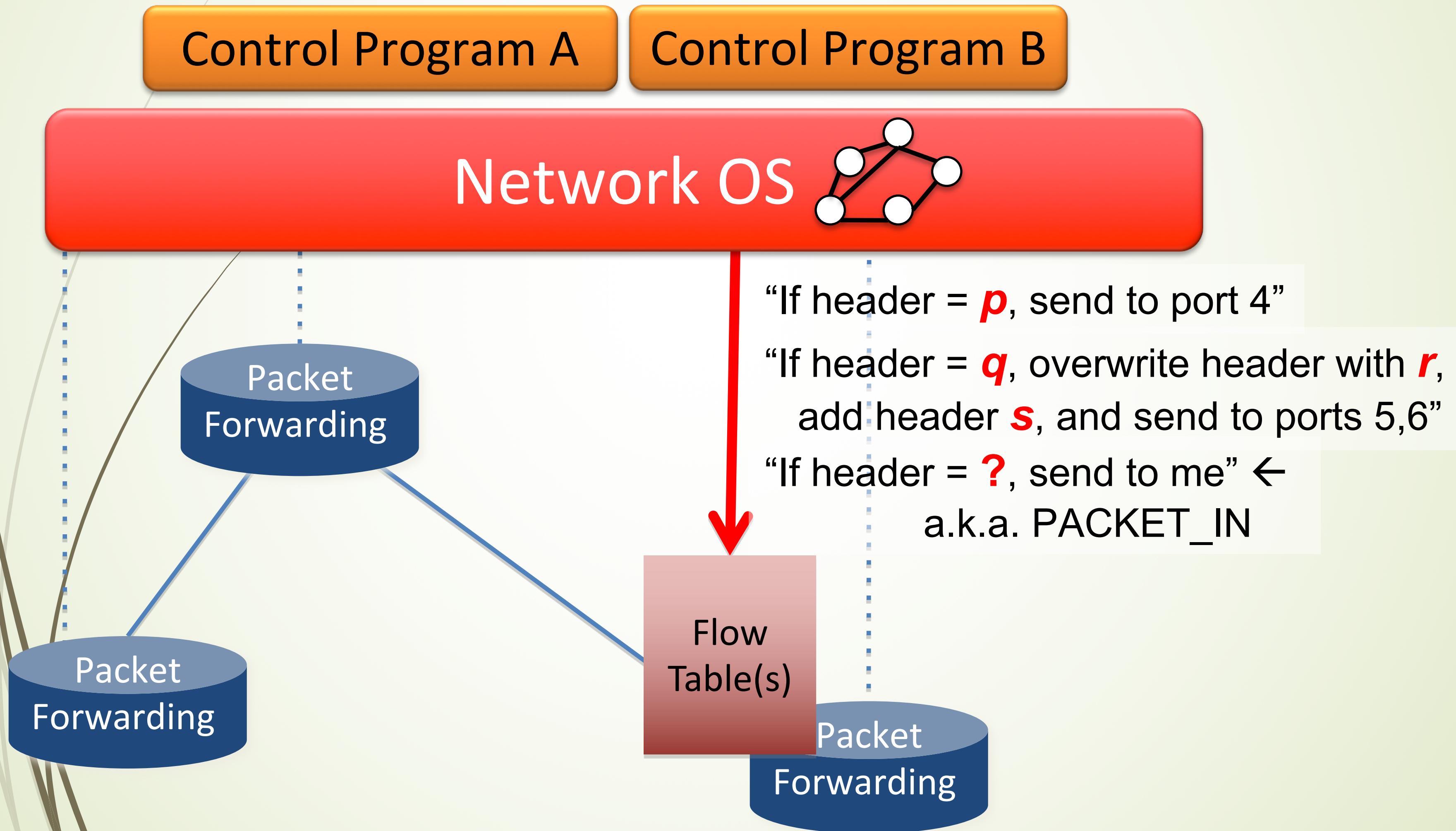
Abstractions in the Control Plane



Data Plane Abstraction

- Purpose: Abstract away forwarding hardware
- Flexible
  - Behavior specified by control plane
  - Built from basic set of forwarding primitives
- Minimal
  - Streamlined for speed and low-power
  - Control program not vendor-specific
- **OpenFlow is a sample protocol of such an abstraction**

OpenFlow in a Nutshell



Agenda

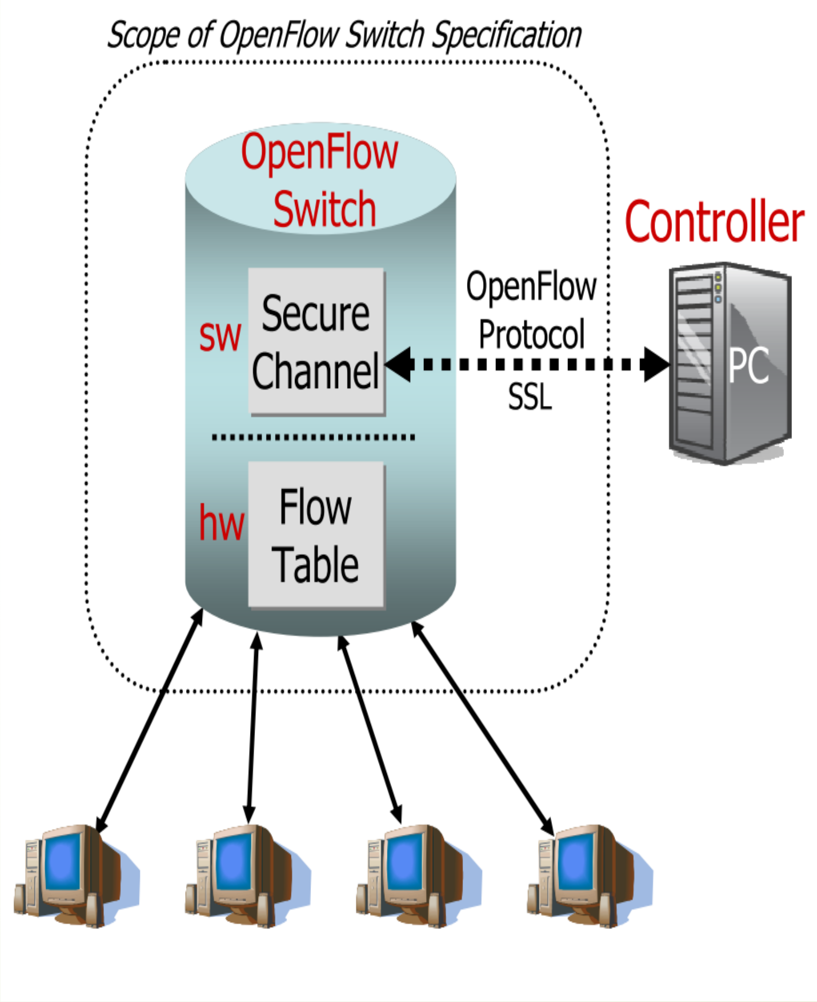
- Motivations
- Concepts
- OpenFlow Protocol
- Network Virtualization
- Virtual Switches

What is OpenFlow

- Similar to an x86 instruction set for the network
  - Provide open interface to “black box” networking node
  - (i.e., Routers, L2/L3 switch) to enable visibility and openness in network
- Separation of control plane and data plane.
  - The data plane of an OpenFlow Switch consists of a Flow Table, and an action associated with each flow entry
  - The control path consists of a controller which programs the flow entry in the flow table
- OpenFlow is based on an Ethernet switch, with an internal flow-table, and a standardized interface to add and remove flow entries

OpenFlow Networks

- Controller (like ONOS)
  - OpenFlow protocol messages
  - Controlled channel
  - Processing
    - Pipeline Processing
    - Packet Matching
    - Instructions & Action Set
- OpenFlow switch (like Open vSwtich)
  - Secure Channel (SC)
  - Flow Table
    - Flow entry



Secure Channel (SC)

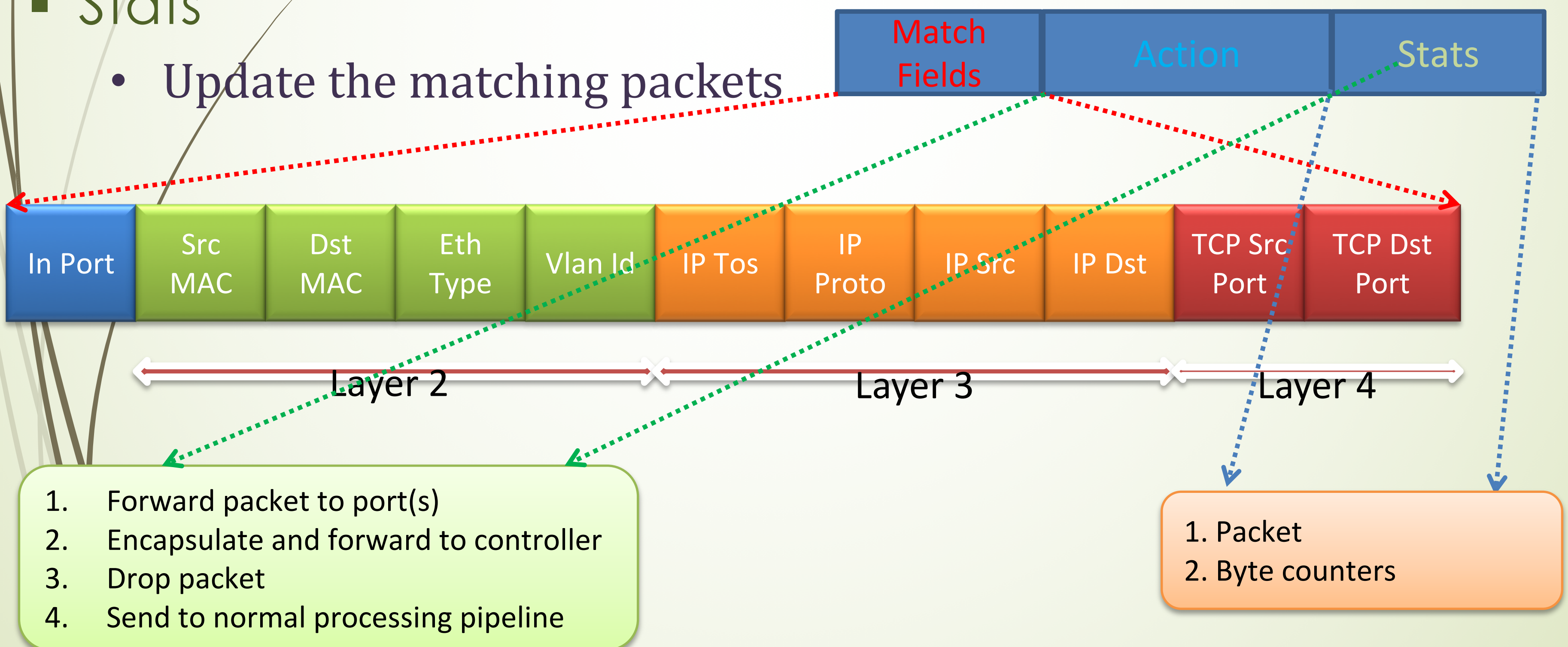
- SC is the interface that connects each OpenFlow switch to controller
- A controller configures and manages the switch via this interface.
  - Receives events from the switch
  - Send packets out the switch
- SC establishes and terminates the connection between OpneFlow Switch and the controller using the procedures
  - Connection Setup
  - Connection Interrupt
- The SC connection is a TLS connection. Switch and controller mutually authenticate by exchanging certificates signed by a site-specific private key.

Flow Table in Switches, Routers, and Chipsets

Flow 1.	Rule (exact & wildcard)	Action	Statistics
Flow 2.	Rule (exact & wildcard)	Action	Statistics
Flow 3.	Rule (exact & wildcard)	Action	Statistics
		....	
Flow N.	Rule (exact & wildcard)	Default Action	Statistics

A Flow Entry Consists of

- Match fields
  - Match against packets
- Action
  - Modify the action set or pipeline processing
- Stats
  - Update the matching packets



Different Usage of Flow Entries

Switching

Switch Port	MAC src	MAC dst	Eth type	VLAN ID	IP Src	IP Dst	IP Prot	TCP sport	TCP dport	Action
*	*	00:1f:..	*	*	*	*	*	*	*	port6

Routing

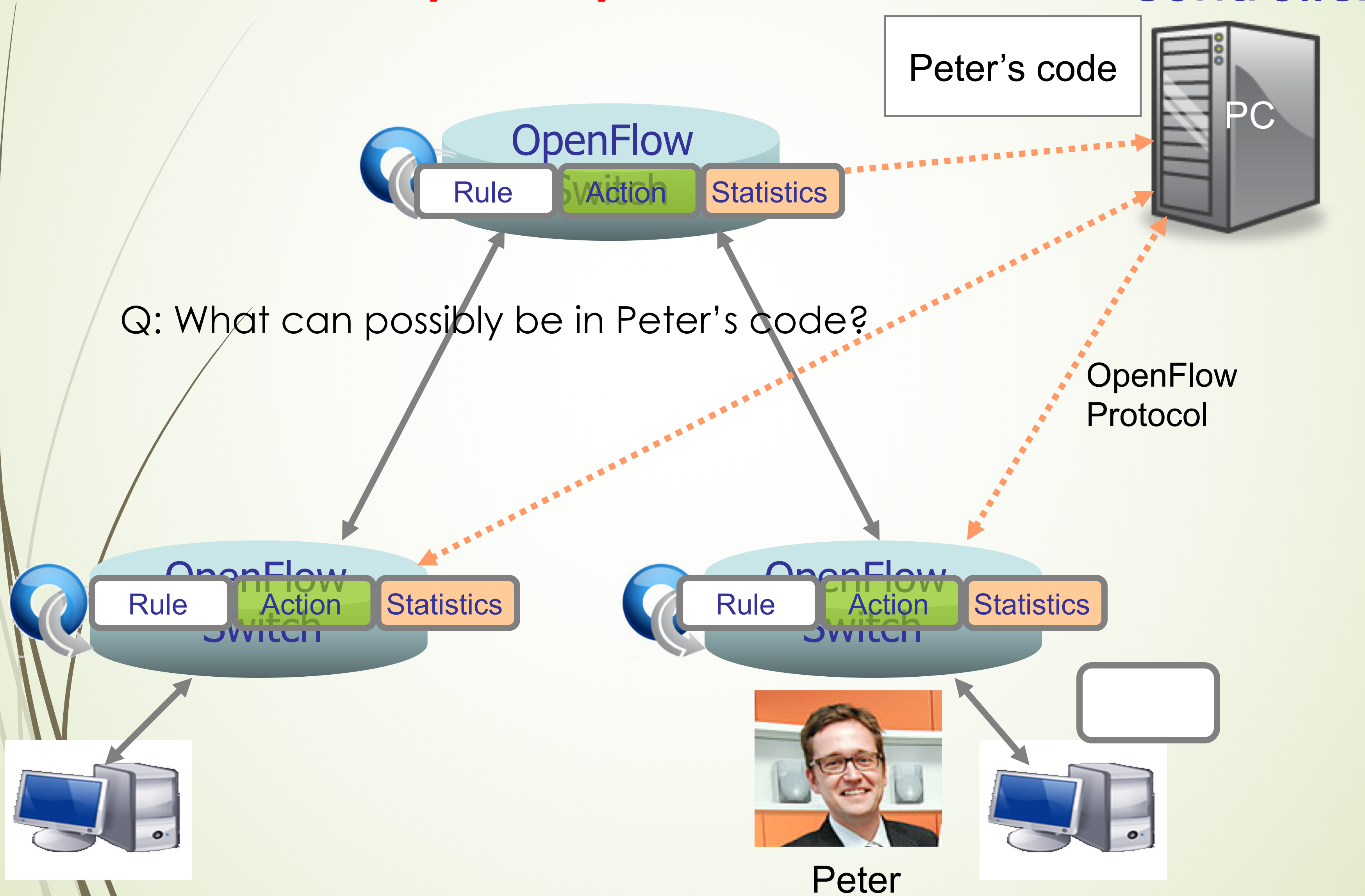
Switch Port	MAC src	MAC dst	Eth type	VLAN ID	IP Src	IP Dst	IP Prot	TCP sport	TCP dport	Action
*	*	*	*	*	*	5.6.7.8	*	*	*	port6

Firewall

Switch Port	MAC src	MAC dst	Eth type	VLAN ID	IP Src	IP Dst	IP Prot	TCP sport	TCP dport	Action
*	*	*	*	*	*	*	*	*	22	drop

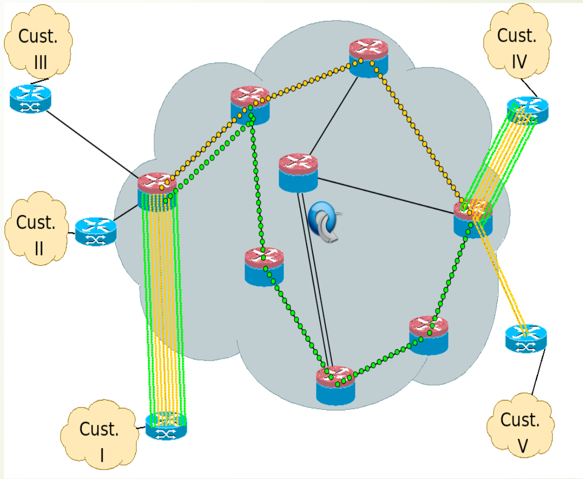
OpenFlow in Actions

Q: What can possibly be in Peter's code?



Sample OpenFlow Application: Aggregation

- Different Networks want different flow granularity (ISP, Backbone,…)
- Current solutions: MPLS, IP aggregation
- OpenFlow-based solution:
  - Dynamically define flow granularity by wildcarding arbitrary header fields
  - Granularity is on the switch flow entries, no packet rewrite nor encapsulation
  - Create meaningful bundles and manage them using your own software (reroute, monitor)



What if we need different controllers?

- **Solution: virtualizing networks!**